

ENERGY MANAGEMENT ENGINEERING HANDBOOK

Volume 03 - Database Design (Code-derived Draft)

This volume was generated by inspecting the uploaded source code. It documents the current SQLAlchemy domain model and intended SaaS database architecture.

Database Hierarchy

Company → Users
Company → Departments
Department → Machine Types
Department → Machines
Machine → Machine Component Instances
Component Type → Component Instances
Component Type → Component Type Tags
Tag Definition → Component Type Tags
Machine Component Instance → Telemetry Data

company.py

Table: company

Columns detected: id, company_name, address

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

component_type.py

Table: component_type

Columns detected: id, name, description

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

component_type_tag.py

Table: component_type_tag

Columns detected: id, component_type_id, tag_definition_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

data_old.py

Table: data

Columns detected: id, timestamp, component_instance_id, tag_definition_id, value

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

department.py

Table: department

Columns detected: id, name, description, company_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

machine.py

Table: machine

Columns detected: id, name, machine_type_id, description, company_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

machine_component_instance.py

Table: machine_component_instance

Columns detected: id, name, component_type_id, machine_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

machine_type.py

Table: machine_type

Columns detected: id, name, description, department_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

tag_definition.py

Table: tag_definition

Columns detected: id, name, unit, description

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

telemetry_data.py

Table: telemetry_data

Columns detected: id, timestamp, component_instance_id, tag_definition_id, value

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

user.py

Table: users

Columns detected: id, username, password_hash, role, company_id

Recommended documentation sections: Purpose, Business Rules, Constraints, Foreign Keys, Indexes, API usage, Example records, Migration strategy.

Next Volume: API and Router Design will be generated from FastAPI routes.